

Draft Appendix: Relativistic Hamiltonian and Scalar Wave Equations

0.1 A Note on the Energy Eigenvalue in the Corrected Schrödinger Equation

The relativistically corrected Schrödinger equation is commonly written as

In particular, the physical time dependence of a stationary relativistic state remains

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \exp\left(-\frac{iE_{\text{tot}}t}{\hbar}\right). \quad (1)$$

Thus, E_{tot} governs the temporal phase, whereas $p_0^2/(2\gamma m_0)$ is the eigenvalue of the equivalent stationary Schrödinger-like equation.

1 Relativistic Hamiltonian and Scalar Wave Equations

This appendix gives the derivation underlying the two scalar refractive-index models used in Section ?? . A complete relativistic description begins with the Dirac equation. When spin and magnetic interactions are not explicitly required, elastic electron diffraction can instead be described by the positive-energy, spin-independent relativistic Hamiltonian [1, 2].

1.1 Minimal Coupling and the Stationary Equation

The time-dependent wave equation is

$$i\hbar \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \hat{H} \Psi(\mathbf{r}, t), \quad (2)$$

with the minimal-coupling Hamiltonian

$$\hat{H} = \sqrt{c^2 (\hat{\mathbf{p}} - q_e \mathbf{A}/c)^2 + m_0^2 c^4} + q_e V(\mathbf{r}), \quad (3)$$

where $\hat{\mathbf{p}} = -i\hbar\nabla$, $\mathbf{A}$ is the magnetic vector potential, and $V(\mathbf{r})$ is the specimen electrostatic potential in volts. We use $e > 0$ and $q_e = -e$, so the electron potential energy is

$$E_{\text{pot}}(\mathbf{r}) = q_e V(\mathbf{r}) = -eV(\mathbf{r}). \quad (4)$$

This notation distinguishes the electrostatic potential V from the energy E_{pot} .

Neglecting magnetic interactions and introducing the rest energy $E_0 = m_0 c^2$ gives

$$\hat{H} = \sqrt{c^2 \hat{\mathbf{p}}^2 + m_0^2 c^4} + E_{\text{pot}}(\mathbf{r}) = \sqrt{c^2 \hat{\mathbf{p}}^2 + E_0^2} + E_{\text{pot}}(\mathbf{r}) \quad (5)$$

We then look for stationary states $\Psi(\mathbf{r}, t)$ that separate the temporal and spatial contributions :

$$\Psi(\mathbf{r}, t) = \psi(\mathbf{r}) \exp\left(-\frac{iE_{\text{tot}}t}{\hbar}\right), \quad (6)$$

The stationary equation is therefore

$$\hat{H}\psi = E_{\text{tot}}\psi. \quad (7)$$

E_{tot} is the conserved total relativistic energy, including the rest energy. As we look for elastic interaction this energy is conserved. In particular, in vacuum,

$$E_{\text{tot}} = \gamma m_0 c^2 = \sqrt{c^2 p_0^2 + E_0^2}, \quad p_0 = \hbar k_0 = \frac{h}{\lambda}, \quad (8)$$

. If the beam has accelerated through a voltage magnitude $U_{\text{acc}} > 0$, the total energy is the sum of the rest and kinetic energies :

$$\text{and } E_{\text{kin}} = eU_{\text{acc}} = E_{\text{tot}} - E_0 = (\gamma - 1)E_0.$$

Equation (8) can be used to compute the wave length : $c^2 p_0^2 = h^2 c^2 / \lambda^2 = E_{\text{tot}}^2 - E_0^2 = (E_{\text{kin}} + E_0)^2 - E_0^2$

1.2 Klein–Gordon and Corrected Schrödinger Forms

Rearranging Eq. (7) and squaring gives

$$[c^2 \hat{\mathbf{p}}^2 + E_0^2] \psi = [E_{\text{tot}} - E_{\text{pot}}(\mathbf{r})]^2 \psi. \quad (9)$$

A local relativistic momentum and a local wavelength can be defined :

$$c^2 \hat{\mathbf{p}}^2 = [E_{\text{tot}} - E_{\text{pot}}(\mathbf{r})]^2 - E_0^2 = \hbar^2 c^2 \mathbf{k}^2 \quad (10)$$

Using $\hat{\mathbf{p}}^2 = -\hbar^2 \nabla^2$ and $p_0 = \hbar k_0$, equation (8) becomes

$$\nabla^2 \psi(\mathbf{r}) + (2\pi k_0)^2 n_{\text{KG}}^2(\mathbf{r}) \psi(\mathbf{r}) = 0, \quad (11)$$

where

$$n_{\text{KG}}^2(\mathbf{r}) = \frac{[E_{\text{tot}} - E_{\text{pot}}(\mathbf{r})]^2 - E_0^2}{E_{\text{tot}}^2 - E_0^2} = \frac{p(\mathbf{r})}{p_0} = \frac{k(\mathbf{r})}{k_0}. \quad (12)$$

So, n_{KG} is the ratio of the local relativistic momentum $p(r)$ to the vacuum momentum p_0 . This connects the Klein–Gordon form to the electron-optical refractive-index concepts of Glaser and Ehrenberg–Siday [3, 4].

For $|E_{\text{pot}}| \ll E_{\text{tot}}$, as it is the case for electron-beam interaction,

$$[E_{\text{tot}} - E_{\text{pot}}(\mathbf{r})]^2 \simeq E_{\text{tot}}^2 - 2E_{\text{tot}} E_{\text{pot}}(\mathbf{r}). \quad (13)$$

Since $E_{\text{tot}}/c^2 = \gamma m_0$, Eq. (9) then reduces to

$$\nabla^2 \psi(\mathbf{r}) + \left[(2\pi k_0)^2 - \frac{2\gamma m_0}{\hbar^2} E_{\text{pot}}(\mathbf{r}) \right] \psi(\mathbf{r}) = 0. \quad (14)$$

or equivalently ,

$$\left[-\frac{\hbar^2}{2\gamma m_0} \nabla^2 + E_{\text{pot}}(\mathbf{r}) \right] \psi(\mathbf{r}) = \frac{p_0^2}{2\gamma m_0} \psi(\mathbf{r}). \quad (15)$$

Equation (15) is therefore a stationary Schrödinger-like reformulation of the relativistic wave equation, rather than the original relativistic energy-eigenvalue equation. This is the relativistically corrected Schrödinger form as deduced by Fujiwara. The right-hand side resembles the nonrelativistic kinetic energy $p_0^2/(2m)$, with the rest mass replaced by the relativistic mass γm_0 . It should not, however, be identified with either the total relativistic energy E_{tot} or, in general, the kinetic energy $E_{\text{kin}} = E_{\text{tot}} - E_0$.

Indeed, using

$$\gamma m_0 = \frac{E_{\text{tot}}}{c^2}, \quad p_0^2 c^2 = E_{\text{tot}}^2 - E_0^2, \quad (16)$$

the eigenvalue in Eq. (15) becomes

$$\frac{p_0^2}{2\gamma m_0} = \frac{E_{\text{tot}}^2 - E_0^2}{2E_{\text{tot}}} = E_{\text{kin}} \frac{E_{\text{tot}} + E_0}{2E_{\text{tot}}}. \quad (17)$$

It approaches E_{kin} only in the nonrelativistic limit $E_{\text{kin}} \ll E_0$, for which $E_{\text{tot}} \simeq E_0$.

Equation xxx-mettrelabonneREF indicates that the energy appearing in the temporal dependence of the wave function is not $E_{0\text{kin}}$ but E_{tot} . The factor 1/2 arises when the scalar relativistic wave equation is linearised in the specimen potential and divided by $2E_{\text{tot}}$.

1.3 Interaction Constant and Paraxial Limit

The conventional TEM interaction constant is

$$\sigma = \frac{2\pi\gamma m_0 e}{h^2 k_0} = \frac{2\pi\gamma m_0 e \lambda}{h^2} = 2\pi e k_0 \frac{\gamma m_0 c^2}{h^2 k_0^2 c^2} = \frac{2\pi e E_{\text{tot}}}{hc \sqrt{E_{\text{tot}}^2 - E_0^2}}. \quad (18)$$

The potential term in Eq. (14) can therefore be written as

$$\frac{2\gamma m_0 e}{\hbar^2} V(\mathbf{r}) = 4\pi k_0 \sigma V(\mathbf{r}), \quad (19)$$

giving

$$\nabla^2 \psi(\mathbf{r}) + [(2\pi k_0)^2 + 4\pi k_0 \sigma V(\mathbf{r})] \psi(\mathbf{r}) = 0. \quad (20)$$

Factoring out the carrier wave,

$$\psi(\mathbf{r}) = \exp(2\pi i k_0 z) \tilde{\psi}(\mathbf{r}), \quad (21)$$

and neglecting $\partial^2 \tilde{\psi} / \partial z^2$ gives

$$\frac{\partial \tilde{\psi}}{\partial z} = \frac{i}{4\pi k_0} \nabla_{\perp}^2 \tilde{\psi} + i\sigma V(\mathbf{r}) \tilde{\psi}. \quad (22)$$

For a slice with projected electrostatic potential

$$V_j(x, y) = \int_{z_j}^{z_j + \Delta z} V(x, y, z) dz, \quad (23)$$

the phase-grating transmission function is

$$t_j(x, y) = \exp [i\sigma V_j(x, y)], \quad (24)$$

which is the expression used in the main text.

References

- [1] K. Fujiwara, [Relativistic dynamical theory of electron diffraction](#), Journal of the Physical Society of Japan 16 (11) (1961) 2226–2238. [doi:10.1143/JPSJ.16.2226](#).
URL <https://journals.jps.jp/doi/10.1143/JPSJ.16.2226>
- [2] E. J. Kirkland, *Advanced Computing in Electron Microscopy*, 2nd Edition, Springer, New York, 2010. [doi:10.1007/978-1-4419-6533-2](#).
- [3] W. Glaser, Über die elektronenoptische abbildung bei hohem Vakuum, Zeitschrift für Physik 83 (1933) 104–122. [doi:10.1007/BF01331099](#).
- [4] W. Ehrenberg, R. E. Siday, The refractive index in electron optics and the principles of dynamics, Proceedings of the Physical Society B 62 (1949) 8–21. [doi:10.1088/0370-1301/62/1/303](#).